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Returning Uluru

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ABSTRACT

Uluru, once known as Ayer's Rock, is one of Australia's most well-known natural phenomena. A World Heritage Site, it is listed for its cultural and natural heritage attributes. Until recently it had been visited by more than 300,000 people annually, many of whom climbed the rock, despite the expressly stated wishes of the Traditional Owners that they should not. In November 2017, the Uluru Kata Tjuta National Park authority decided to permanently close the Uluru climb and remove the climb infrastructure, which included a series of 134 posts cemented into the rock and linked by chain, with a cairn at the summit. This paper details the conservation processes and remediating materials considered, then the methods employed to remove this infrastructure safely. Technical requirements included leaving no visible evidence of the posts, avoiding or minimising any damage to the rock surface, and providing a long-lasting in-fill solution that would blend into the rock surface, on the basis that the rock would never be climbed again. The logistics of a site which is some 470 km from the nearest town, has no facilities, and where the surface temperature can reach 60–70°C in mid summer, determined the removal of the evidence of human intervention on the rock site of Uluru. The broader context of Uluru's recent history and its hand-back to the traditional owners is also discussed.

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Uluru; monument conservation; natural site conservation; world heritage; traditional owners

Introduction

This paper describes arguably Australia's most well-known natural wonder, and also one of its most sacred sites. Uluru is a monolith of arkosic sandstone, a mixture of quartz and orthoclase feldspar, set amidst a flat plain of heavy soils in Australia's Northern Territory. It is about 9.4 kilometres in circumference, rising to a height at its summit of 340 metres above the surrounding plain (Figure 1).

In 1873 the explorer William Gosse became the first non-Aboriginal person to see Uluru, naming it Ayers Rock after the Chief Secretary of South Australia, Sir Henry Ayers. The monolith was known by this name until the creation of the Uluru Kata Tjuta National Park (UKTNP) in 1985. The UKTNP is listed under the World Heritage Convention[1] for both its cultural and natural heritage attributes and values, meeting four criteria:

- an outstanding example representing significant on-going geological processes, biological evolution and man's interaction with the natural environment
- contains unique, rare or superlative natural phenomena, formations or features or areas of exceptional natural beauty, such as superlative examples of important ecosystems to man, natural features, sweeping vistas covered by natural vegetation and exceptional combinations of natural and cultural elements
- a cultural landscape representing the combined work of nature and of man, manifesting the interaction between humankind and its natural environment
- an associative landscape having powerful religious, artistic and cultural associations of the natural element.

Local Aboriginal people refer to themselves as *Anangu* and their belief system as *Tjukurpa*. Aboriginal people have lived in the area around Uluru and the Kata Tjuta region for at least 30,000 years. For *Anangu*, their culture has always existed here: the central Australian landscape (of which Uluru and Kata Tjuta are an important part) is believed to have been created at the beginning of time by ancestral beings, and Uluru provides physical evidence of feats performed during the creation period, which are told in the *Tjukurpa* stories. *Anangu* are the direct descendants of these beings and are responsible for the on-going protection and appropriate management of these ancestral lands.

Uluru has been a famous tourist destination attracting some 300,000 visitors a year despite the harshness of the surrounding terrain and the nearest town (Alice Springs) being 467 km distant. Many visitors have chosen to climb the rock, despite clear requests not to do so from the Traditional Owners since the creation of the national park. There was immense pressure from the tourism industry to maintain the climb as an important economic contribution to the Northern Territory.

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Figure 1. Overall view of Uluru from the air. Image: David Wallis.

The visitor climb up to the top of Uluru is steep and strenuous and the rock surface along the first part of Uluru is relatively smooth, due to footwear from high numbers of visitors having climbed over several decades. To make the climb safer after a number of deaths occurred due to falls, climb posts were installed in two stages in the mid-1960s and 1970s, with a chain providing a secure safety line for visitors to hold onto while climbing up and down Uluru (Figure 2), as well as a cairn on the highest point. Despite this, a number of visitor deaths still occurred over the years.

The site is managed by both the traditions of the *Anangu* (traditional owners) and western management practices and laws upheld by the UKTNP authority. On 1 November 2017, the UKTNP Board unanimously decided, in accordance with policy 6.3.3 of the UKTNP Management Plan, to permanently close the Uluru climb, with the closure to commence from 26 October 2019. This followed various preconditions being met, namely a drop in the proportion of those visitors wishing to climb Uluru to below 20%, and evidence that cultural and natural experiences had become critical factors influencing visitors to the national park.

The importance of this action to the traditional owners is best understood by a statement made by Sammy Wilson, chair of the board of management of UKTNP:[2]

> *Anangu* have always held this place of Law. Other people have found it hard to understand what this means; they can't see it. But for *Anangu* it is indisputable. So this climb issue has been widely discussed, including by many who have long since passed away. More recently people have come together to focus on it again and it was decided to take it to a broader group of *Anangu*. They declared it should be closed. This is a sacred place restricted by law.
>
> It's not just at board meetings that we discussed this but it's been talked about over many a camp fire, out hunting, waiting for the kangaroo to cook, they've always talked about it.
>
> The climb is a men's sacred area. The men have closed it. It has cultural significance that includes certain restrictions and so this is as much as we can say. If you ask, you know they can't tell you, except to say it has been closed for cultural reasons.
>
> What does this mean? You know it can be hard to understand – what is cultural law? Which one are you talking about? It exists; both historically and today. *Tjukurpa* includes everything: the trees; grasses; landforms; hills; rocks and all.
>
> You have to think in these terms; to understand that country has meaning that needs to be respected. If you walk around here you will learn this and understand. If you climb you won't be able to. What are you learning? This is why *Tjukurpa* exists. We can't control everything you do but if you walk around here you will start to understand us.
>
> The land has law and culture. We welcome tourists here. Closing the climb is not something to feel upset about but a cause for celebration.

Following the UKTNP board's decision to close the Uluru climb, the board approved a strategy to implement its permanent closure.[3] That strategy set out actions to implement the closure of the Uluru



Figure 2. Climb path prior to removal. Image: David Wallis.

climb, one such action being to 'Remove Uluru climb walking route facilities and permanently close the climb'.

Climb infrastructure

The posts and chain were installed in two separate stages, the first in 1966 and the second in 1976. The specifications for the chain and the posts were calculated by John Ryan from the Darwin office of Guthrie Haskins and Davy (now GHD). Ryan based his calculations on the chain being able to withstand a 125 kg man holding onto the chain at an angle of 45°. Both stages were installed by Alan White, a local electrician. A detailed description of the chain installation survives through the transcript of an interview conducted by Linda Rive at the Uluru Cultural Centre in 1998 with White.[4]

There were 138 steel posts drilled and fixed into Uluru. The first post was about 100 metres in linear distance from the ground surface at the base of Uluru. On this first/main section of posts and chain there were 136 posts and a continuous section of chain over a linear distance of about 450 metres. There were two separate posts just above (and about five metres away from) the main section of the posts/chain. The steel posts were about 42 mm in diameter (with a wall thickness of approximately 3 mm) and an above ground height of about 700 mm. Posts were an average distance of approximately 3.2 m apart. There were two different sizes of chain, one made of steel approximately 8 mm in diameter and one made of steel approximately 13 mm in diameter, and each post had a section of chain bolted to it. The posts were drilled into the rock surface to a depth of approximately 300 mm and bonded to the rock with an epoxy resin, with each post also filled with concrete/epoxy resin known as Expandite.

On top of the summit of Uluru a cairn was formed of stones cemented together in an octagonal shape and

painted a reddish brown colour to blend in with the colour of Uluru. The cairn was about 90 cm in height and about 70 cm in diameter with a brass direction dial on the top of it, with the distances to significant land marks visible in the region and more broadly across Australia marked on it.

Planning for the removal of the infrastructure

Following the decision by the UKNTP board to close the climb and remove its infrastructure, a tender was called for the removal of the chain, posts and cairn, with the following requirements:

- the removal of the Uluru climb facilities must not leave any visible evidence of the posts above the natural rock surface of Uluru
- the removal of the Uluru climb facilities must avoid or minimise any damage to the rock surface of Uluru
- no part of the steel posts can be left exposed above the surface of the rock
- the fill in the remaining holes must be durable and long lasting, and be level with and blend into the existing rock surface of Uluru (in terms of its texture and colour).

Various issues needed to be carefully considered prior to the implementation of the removal. The work could not begin before the Central Land Council completed consultation with the traditional owners. A sacred site clearance certificate had to be produced to authorise the works, followed by the handback of Uluru to the Traditional Owners on 26 October 2019. This meant that the works had to be undertaken in early to mid southern hemisphere summer (October 2019–December 2019) when typical mean daily shade temperatures range from 20 to 38°C, but are significantly higher in direct sun, and higher again on Uluru due to reradiated heat from the rock surface. They are in the range of 60–70°C by mid-morning. These temperatures are well in excess of recommended guidance for the use of cement mortars and organic adhesives such as epoxy, because they could have significant deleterious impact on their workability, curing and long-term performance. In addition, such high temperatures introduce workplace health and safety issues, potentially impacting on the concentration of the workers carrying out this project as well as related logistical issues.

Removal of the steel pipe rather than cutting was seen as preferable, and various methods were considered for this. These included applying lateral pressure with a pipe wrench, upward pressure with a hollow jack or downward pressure with a sledge hammer, the purpose of each being to break the bond between the metal pipe and its cementitious/epoxy grout.

However if the posts could not be safely removed without damaging the surrounding surface of the rock the implication of leaving the posts in the rock required careful consideration. The posts were galvanised steel pipe, the zinc galvanisation providing protection against corrosion of the steel alloy. Once cut, the exposed surface of the steel would be more susceptible to future corrosion through the formation of rust (iron oxide). As corrosion progresses, the layers of rust expand; as a rule of thumb, the corrosion products occupy ten times the original volume of steel. Where steel is embedded in other materials, this leads to high stresses in the other materials, resulting in both cracking and displacement of surrounding material by spalling. The rate of future corrosion of any galvanised steel pipe left embedded below the surface of Uluru could be anticipated to be low due to the comparatively benign environment for corrosion: the site has low annual rainfall, and though it has high temperatures, there is low relative humidity at all times.

However it was identified that mitigation of the risk arising from leaving the galvanised steel pipe embedded beneath the surface of Uluru should include application of an appropriate film thickness of a zinc-rich epoxy primer.

It was also identified that the process of cutting or grinding the chain or posts would create grinder swarf which would need careful collection to avoid any 'rust-staining' of the surface of Uluru, even though its visibility would be low, given the existing strong orange colour attributable to iron oxide already present in the sandstone forming Uluru.

Drawing on best practice conservation stonemasonry techniques for patching damage to stone in buildings, two broad approaches to the filling of holes were identified:

- patching with 'artificial stone', either mortar-based (cementitious or lime-based) or resin-based (epoxy or polyester), with a variety of aggregate fillers
- indenting with matching natural stone.

The development and use of resin-based 'artificial stone' patching techniques has accompanied the rise of such materials in other construction use over the past 40–50 years. There are two major strands of resin-based stone patching – one utilises polyester resins and is generally used for factory-based patching of low-porosity stones such as marble and granite, although some site-installed patching using these materials does occur. The other strand involves the mixing of epoxy resins with crushed stone (limestone or marble dust) or sand, to achieve a colour and texture match for more porous stones such as sandstone or limestone.

The performance of resin-based patches in a range of different stone materials in buildings in Australia suggests that the service life varies between 10 and 30 years depending on the resin, the resin-filler ratio, and the exposure to ultraviolet (UV) radiation. For this project, the use of a resin-based patching material was identified as high risk, given the extremely high exposure to UV outdoors, and the very substantial diurnal and seasonal temperature variations on the surface of Uluru.

The use of lime- or cement-based mortars for patching of porous stones such as sandstone is relatively widespread. There has been a tendency in historic building circles to move away from cement-based patching mortars toward lime-based mortars, due to a perception that cementitious materials are more deleterious when in contact with stone.

Both lime-based and cement-based patching mortars have the potential for leaching of soluble salts into the surrounding stone, with the consequent possibility of accelerated deterioration of the surrounding stone. This is less likely in arkosic sandstones such as the Mutitjulu Arkose which forms the majority of Uluru than in the argillaceous sandstones widely used in buildings in the major cities of Australia, where such mortars have been employed.

Whilst both lime-based and cement-based mortars require curing, lime-based mortars tend to be more sensitive to curing conditions, and require the presence of moisture for a longer period of time. On this basis, a cementitious mortar mix was identified as more appropriate than a lime-based mortar mix.

The use of carefully shaped natural stone pieces indented into the surface of existing stone blocks in buildings is a well-established method of patching by stonemasons. The area of damage is generally chiselled or otherwise cut to a regular shape which can be matched during fabrication of the indent piece. The piece of stone to be indented is then generally bonded into place using an adhesive method (either cementitious or epoxy-based). Standard stonemasonry practice for indents is to specify a 'hairline' joint, which will typically be less than 1 mm in dimension, in order to minimise the visual impact of the indent repair.

In this instance, two options for indents were considered.

- A piece of stone measuring 20–50 mm in depth (depending on the depth to which the steel post-base had been removed) can be fabricated and adhered into position. These indent pieces could be prepared from loose rock which had separated from the Uluru monolith, either from the top of the monolith, or from around the perimeter at the base.
- A cylinder of stone to the full depth of the hole could be installed (assuming that the entire post has been removed), with adhesive inserted around the top 30–100 mm of the perimeter joint with the parent rock. This approach would require preparation of at least 150 cylinders, by coring matching rock material away from the site.

All these issues were considered at length and trials were undertaken off-site before the on-site part of the project started.

Planning the on-site process

The site work would begin with the removal of the cairn. This required careful evaluation of its current condition and construction, along with the constraints of transport by helicopter.

The sensitivity of the project made a small footprint to the works desirable. To this end, a four-person crew working with cordless power tools and light rope-based rigging systems was decided upon, with limited helicopter support to aid in the removal of the estimated 3000 kg of materials from the climb infrastructure.

The 400 m long handrail chain and its accompanying posts were proposed to be removed starting from the bottom of the former climb and working upwards. This 'bottom up' removal had a twofold benefit; it allowed the remaining 'upslope' handrail posts to form the basis of an anchor system for worker safety lines, whilst the infrastructure disappearing from the bottom of the climb and up out of sight would dissuade any would-be trespassers from attempting the climb after the closure. Furthermore, the work location and equipment on-site would be protected from the public.

The extreme heat was problematic, and not only for the safety and productivity of the crew. It also had the potential to reduce the lifting capacity of the helicopters used for logistical support by as much as 30%.

A hollow-bore hydraulic cylinder was used in the off-site trials to successfully remove chemically fixed 42 mm diameter steel tube from a concrete substrate, but as it was capable of exerting a 60 ton jacking force it would have caused compressive damage to the rock surface before any outward movement of the posts occurred. As a consequence, cutting the posts, then recessing the remaining steel below the surface, was chosen as the removal method.

In order to make the removal of materials as efficient as possible and to reduce manual handling and foot traffic on the former climb, tensioned rope highlines, elevated above the surface of the rock by aluminium A-frames, were rigged along the line of the chain. This allowed the removed materials to be belayed down the highlines and directly into the tray of a

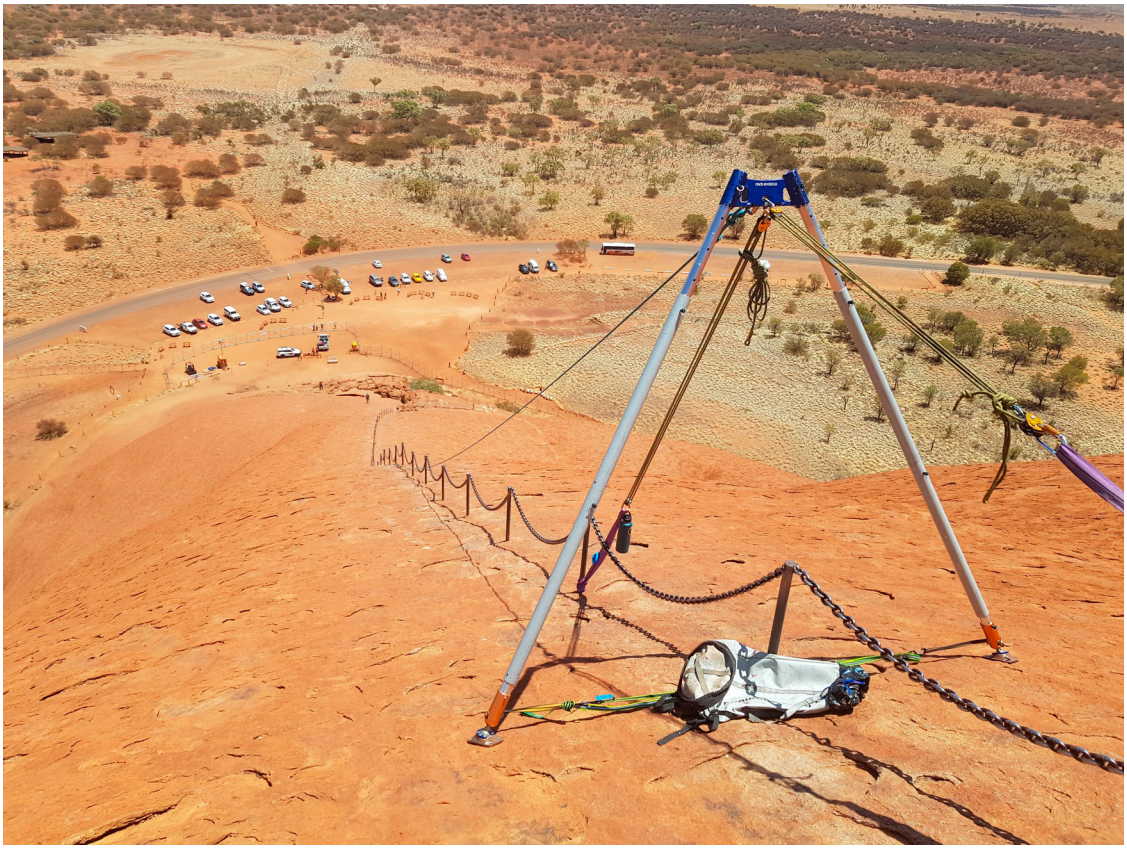


Figure 3. Climb path with A frame. Image: David Wallis.

truck waiting on the ground. The highlines also served as a rescue system to allow the evacuation of an injured worker in the event of an emergency.

The removal process

The crew of four split up for the first phase of the works. Two headed to the summit (some 350 m above the surrounding plain and around 1600 m in horizontal distance from the base) to dismantle the summit cairn and remediate the site, whilst the other two rigged the rope highline system on the initial 400 m of the climb.

Construction of the cairn was found to be exactly as depicted in documents obtained from a National Archives release. Consequently, disassembly could be targeted. The facing stones of the summit cairn were numbered according to position and orientation prior to disassembly. Each face of the cairn was photographed, showing the numbered stone positions.

A diamond blade grinder and light duty mechanical breaker were then employed to cut the cementitious mortar from around the facing stones under the plaque. Behind the facing stones was a fill of stone rubble poorly bonded together with mortar. The base layer of mortar had limited adherence to the rock substrate and was easily removed without impact. The facing stones and stone rubble fill were separated from the mortar and loaded into separate 'bulker' bags. The loads were weighed as the bags were filled (for accurate reporting of load sizes to the helicopter pilot). The three sleeve anchors fixing the plaque, *via* its pedestal, to the rock surface were removed, and the holes were filled with a natural stone and resin mix.

The bulker bags were wrapped in netting and along with the plaque, flown off the summit the following morning, in two lifts *via* helicopter, then returned to the park workshop (2 km south-west of Uluru) where the stonemason commenced reassembly of the cairn using fresh mortar.

Simultaneously, the climb handrail chain was unbolted from the posts and the chain lengths were 'broken' at intermediate hammerlock-style joining links. This allowed the retention of complete lengths of chain for display purposes and limited the need for cutting. The chain was then strung to the highline *via* a series of pulleys and karabiners and lowered down in approximately 15 m sections.

The handrail posts, 42 mm diameter steel tube fixed into the rock using Expandite cementitious grout (for the 1960s posts) or an unidentified resinous product (In the case of the 1970s posts), had been core-filled and embedded to a depth believed to be 450 mm. The ideal of achieving complete removal of the posts had to be tempered with the absolute requirement to cause no further damage to the rock itself during

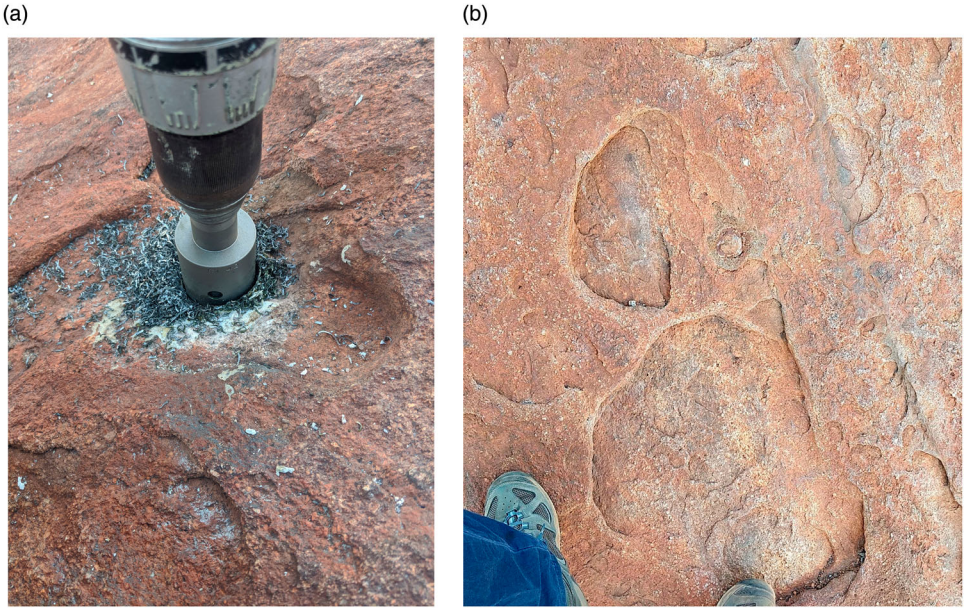


Figure 4. (a) Milling post base below surface. (b) Remediated hole. Images: David Wallis

the process. One of the traits of Uluru's arkose sandstone is the formation of surface exfoliation flakes. Accordingly, even along the climb path which had seen the passage of many thousands of walkers, much of the surface was hollow-sounding and delaminated.

Tools were loaded into a large capacity plastic case mounted to a wheeled trolley. The trolley was hauled uphill from the carpark using the highline system (Figure 3) until the lowest handrail post was reached, thereafter it was towed uphill from one post to the next using a compact 4:1 pulley system.

Each post was first cut as close as possible to its base using a 23 cm cordless grinder fitted with a diamond cutting wheel. The mortar or resin core fill of the base was then removed using a rotary hammer drill with a 35 mm masonry bit. 42 mm carbide-toothed hole saws designed for coring holes in plate steel were employed to recess the bases below substrate surface level. In workshop testing, the hole saw pilot bit was replaced with a turned block of Delrin which matched the inside diameter of the Uluru climb posts. This turned block kept the carbide teeth centred over the wall of the post and enabled up to 50 mm of post length to be removed in five minutes of drilling.

Twenty-five millimetre nominal bore steel pipe was found to have an outside diameter that was just less than the inside diameter of the Uluru handrail posts so, for longevity in the field, the hole saw guides were changed to this.

All 140 post bases were milled down a minimum of 20 mm below the rock surface (Figure 4(a)). All swarf generated by the drilling and the dust from removing the post cores was vacuumed up and disposed of off-site. The holes left by the removal of the posts then had a bed of Chemset polyester resin laid in them and were then filled with natural stone (Figure 4(b)).

When park officials were approached for suitable sources of natural stone to fill the post holes, they gave access to their store of 'sorry rocks'. Uluru has a long history of visitors removing small pieces of rock for souvenirs, during their stay. Whether through a sense of guilt or fear of bringing on bad luck, these rocks are often returned to the park in the post, generally accompanied by a note of explanation. They have become known as 'sorry rocks'. The 'sorry rocks' were shaped using a scutch hammer to fit the individual holes. Their use provided a both practical and symbolic solution to the desire to use natural stone to repair the rock surface.

At the end of this process, the toolbox, safety lines and final bags of cut posts, along with the work party, were lifted off *via* helicopter, and demobilisation of the work site was completed. All materials removed from the climb, along with the restored summit cairn, have been put into storage so that a decision can be made about the best way to display them in the future. A small number of brass memorial plaques at the base of Uluru were also removed for repatriation to families of deceased persons.

Conclusion

In May 2019, 1300 Aboriginal and Torres Strait Islander people from all parts of Australia came together to develop a path forward for enshrining indigenous Australians in the nation's constitution. The place they chose to meet was Uluru, as an iconic spiritual place in the centre of Australia. The outcome of this meeting was the *Uluru Statement from the Heart*. The



Figure 5. Climb path after removal. Image: David Wallis.

importance of Uluru to its traditional owners was once again strongly emphasised. This removal of the chain and posts and the cairn was a highly important one in what it achieved physically but also what it meant spiritually. The way in which it was approached was therefore critical to the overall success of the project.

The project was undertaken in a collaborative fashion between the Traditional Owners, the UKTNP, consultants and contractors. It successfully considered the cultural, logistical and practical requirements. The evidence of human and tourist activity which had taken place on the rock was manifested by the chain posts and cairn but also included the piling of loose rocks into makeshift cairns, graffiti (scratching and painting names into the surface of Uluru), defaecating and littering. That evidence of what took place over a brief 150 years in the rock's history will soon be erased by weathering, and Uluru, the spiritual heart of Australia, will lie undisturbed once more (Figure 5).

Notes

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Disclosure statement

No potential conflict of interest was reported by the author(s).